

Dr. Virendra Ashiwal

Senior Research Scientist — Applied AI & Industrial Automation

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Profile

Senior Research Scientist and Software Architect with over a decade of combined industrial and academic experience at the intersection of applied artificial intelligence, multi-agent systems and distributed control. Leads corporate R&D initiatives connecting generative-AI research with production-oriented industrial systems. Research spans LLM-assisted control engineering, agentic automation, industrial cybersecurity and modular IEC 61499 architectures.

Core Expertise

Research leadership: Applied-AI strategy, R&D project management, cross-functional teams, technology roadmaps, academic-industrial collaboration
AI and architecture: Multi-agent systems, LangGraph, CrewAI, generative AI, RAG, LLMs for code and testing, distributed systems, microservices, Apache Kafka
Industrial domain: PLC/DCS software architecture, IEC 61499, IEC 61131-3, Eclipse 4diac, APROL, OPC UA, MQTT, motion control, drive commissioning

Professional and Research Experience

- Senior Research Scientist & Project Lead

ABB Corporate Research Center

10/2025–Present

Mannheim, Germany

 - Lead strategic applied-AI research focusing on generative AI, multi-agent frameworks and intelligent industrial automation.
 - Direct cross-functional R&D teams connecting machine-learning research with robust industrial systems and drive applications.
- Research Scientist

ABB Corporate Research Center

11/2022–Present

Ladenburg, Germany

 - Spearheaded the architecture and delivery of DriveAIAgent, a multi-agent system for industrial drive commissioning and troubleshooting.
 - Research LLM-based engineering, industrial cybersecurity, on-premise analytics and knowledge-grounded automation workflows.
- Lead Researcher / PhD Candidate

LIT Cyber-Physical Systems Lab, Johannes Kepler University

04/2019–10/2022

Linz, Austria

 - Conceived and developed the PLC-Service Bus: an event-based modular architecture using IEC 61499, Apache Kafka and OPC UA to reduce rigid control dependencies.
 - Conducted doctoral research and academic-industrial collaboration; contributed to IEEE ETFA as program committee member, reviewer and session chair.
- Technical Expert — Process & Factory Automation

B&R Industrial Automation GmbH

11/2015–02/2019

Eggelsberg, Austria

 - Delivered global second-level support, business-development consulting and proof-of-concept systems for Industry 4.0 applications across European and Asian markets.
- Application Engineer

B&R Industrial Automation

07/2011–07/2012

Delhi, India

 - Engineered automation and control infrastructure, multi-axis servo solutions and industrial communications.

Education

PhD in Computer Science, Johannes Kepler University Linz2026

Dissertation: PLC-Service Bus: An Architecture and Development Framework for Modular and Adaptable PLC Software; Supervisor: Prof. Dr. Alois Zoitl.

Selected Publications

- H. Kozirolek, T. Braun, **V. Ashiwal**, et al. “Spec2Control: Automating PLC/DCS control-logic engineering from natural language requirements with LLMs—a multi-plant evaluation.” *IEEE/ACM ICSE SEIP*, 2026.
- **V. Ashiwal**, M. Ritter, S. Palacio, N. Schoch. “DriveAI Agent: A Multi-Agent System for Industrial Drive Commissioning and Troubleshooting.” *IEEE ETFA*, 2025.
- **V. Ashiwal**, O. Miguel-Escrig, B. Wiesmayr, A. Zoitl, J.-A. Romero-Pérez. “Identification and Evaluation of Pitfalls in the Migration from IEC 61131-3 to IEC 61499: A Review.” *IEEE OJIES*, 2025.
- H. Kozirolek, S. Grüner, R. Hark, **V. Ashiwal**, et al. “LLM-Based and Retrieval-Augmented Control Code Generation.” *LLM4Code@ICSE*, 2024.
- H. Kozirolek, **V. Ashiwal**, S. Bandyopadhyay, C. K. R. “Automated Control Logic Test Case Generation Using Large Language Models.” *IEEE ETFA*, 2024.
- **V. Ashiwal**, S. Finster, A. Dawoud. “LLM-Based Vulnerability Sourcing from Unstructured Data.” *IEEE EuroS&PW*, 2024.
- H. Kozirolek, S. Gruener, **V. Ashiwal**. “ChatGPT for PLC/DCS Control Logic Generation.” *IEEE ETFA*, 2023.
- **V. Ashiwal**, A. M. Gutierrez, K. Aschbacher, A. Zoitl. “Apache Kafka as a Middleware to Support the PLC-Service Bus Architecture with IEC 61499.” *ECSA*, 2022.
- **V. Ashiwal**, A. Zoitl, M. Konnerth. “A Service Bus Concept for Modular and Adaptable PLC-Software.” *IEEE ETFA*, 2020.

Selected Patent Applications

- N. Schoch, M. Elsheikh, **V. Ashiwal**. “System and Method Based on a Group of LLM-Based Agents for Generation and Enhancement of Engineering-Data-Funnel Outputs.” US App. 19/332,302, 2026.
- **V. Ashiwal**, N. Schoch, P. Becker. “Method for Obtaining an AI Agent, Methods for Usage of Said AI Agent, Control Apparatus, Automation System...” US App. 19/083,604, 2025.
- C. Song, **V. Ashiwal**, P. Becker, N. Schoch. “Method for Obtaining Domain-Informed ML/AI Model...” US App. 19/083,571, 2025.
- S. Gruener, H. Kozirolek, **V. Ashiwal**. “Method for Generating a Control Logic Code for Controlling an Automated Industrial Process.” US App. 19/086,527, 2025.

Research Service, Certifications and Languages

Service: IEEE ETFA Track 1 Program Committee and reviewer (2021–Present); peer reviewer for IEEE OJIES, CASE and INDIN; IEEE ETFA session chair (2021).

Certifications: Professional Scrum Product Owner II; iSAQB Certified Professional for Software Architecture — Foundation Level; LangChain and LangGraph Essentials.

Languages: Hindi (native/bilingual), English (full professional), German (B1 certified).